

SMI CRL microfocusing: initial feasibility

PROVISIONAL OPTICS MODEL - September 2026

Lens plane to sample at $Z=0$: 615 mm

Z travel: -50 to 250 mm; positive downstream

Accessible image distance: 365 to 665 mm

Incident curvature at $Z=0$: 0 /m

Eight populated holders; 255 nonempty combinations evaluated.

Priority: fewest holders, then elements, then holder numbers.

Predicted gaps within 2.1-24 keV (analytic boundaries):

2.6052 - 2.9180 keV

ASSUMPTIONS

Diamond, density 3.515 g/cm³; two parabolic surfaces per lens.

Quoted radii apply to each surface. All holders share one plane.

Free-electron δ (carbon $f_1=6$); δ scales as energy⁻².

At 10 keV: $\delta = 7.29009e-06$.

Focal length: $f = 1 / [2 \delta \sum(N/R)]$.

Ray curvature C: $1/q = 1/f - C(Z)$; $q = (615-Z)$ mm by default.

$C(Z) = C(0) / [1 + C(0) Z(m)]$. Positive C means divergence.

INTERPRETATION

Coverage means a geometrical focus exists within nominal travel.

It does not establish spot size, transmission, or beam quality.

Material, radius convention, holder spacing and Z zero need confirmation. No tabulated dispersion correction is included.

Transmission needs lens aperture and web thickness; spot size also needs incident size/emittance and wavefront information.

Fewest holders can favor a thick stack over fewer lens elements.

Switches may require large Z jumps. Boundary solutions have zero travel margin; usable coverage will shrink with a margin.

A 1% focal-length error shifts Z by roughly 3.7-6.7 mm here.

Model and report generation issue no hardware commands.

Holder inventory and nominal positions

Holder	Contents	IN mm	OUT mm
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1	1 x 50 um (assumed diamond)	0.0	10.0
2	8 x 50 um (assumed diamond)	0.0	10.0
3	16 x 50 um (assumed diamond)	0.0	10.0
4	4 x 50 um (assumed diamond)	0.0	10.0
5	2 x 50 um (assumed diamond)	0.0	10.0
6	empty	0.0	10.0
7	empty	0.0	-10.0
8	empty	0.0	-10.0
9	1 x 200 um (assumed diamond)	0.0	-10.0
10	8 x 500 um (assumed diamond)	0.0	-10.0
11	1 x 500 um (assumed diamond)	0.0	-10.0
12	2 mm aperture	0.0	-10.0

Empty holders 6-8 and aperture holder 12 have no lens power.

They are excluded from automatic lens selection.

IN=0 mm is nominal, pending alignment within about +/-2 mm.

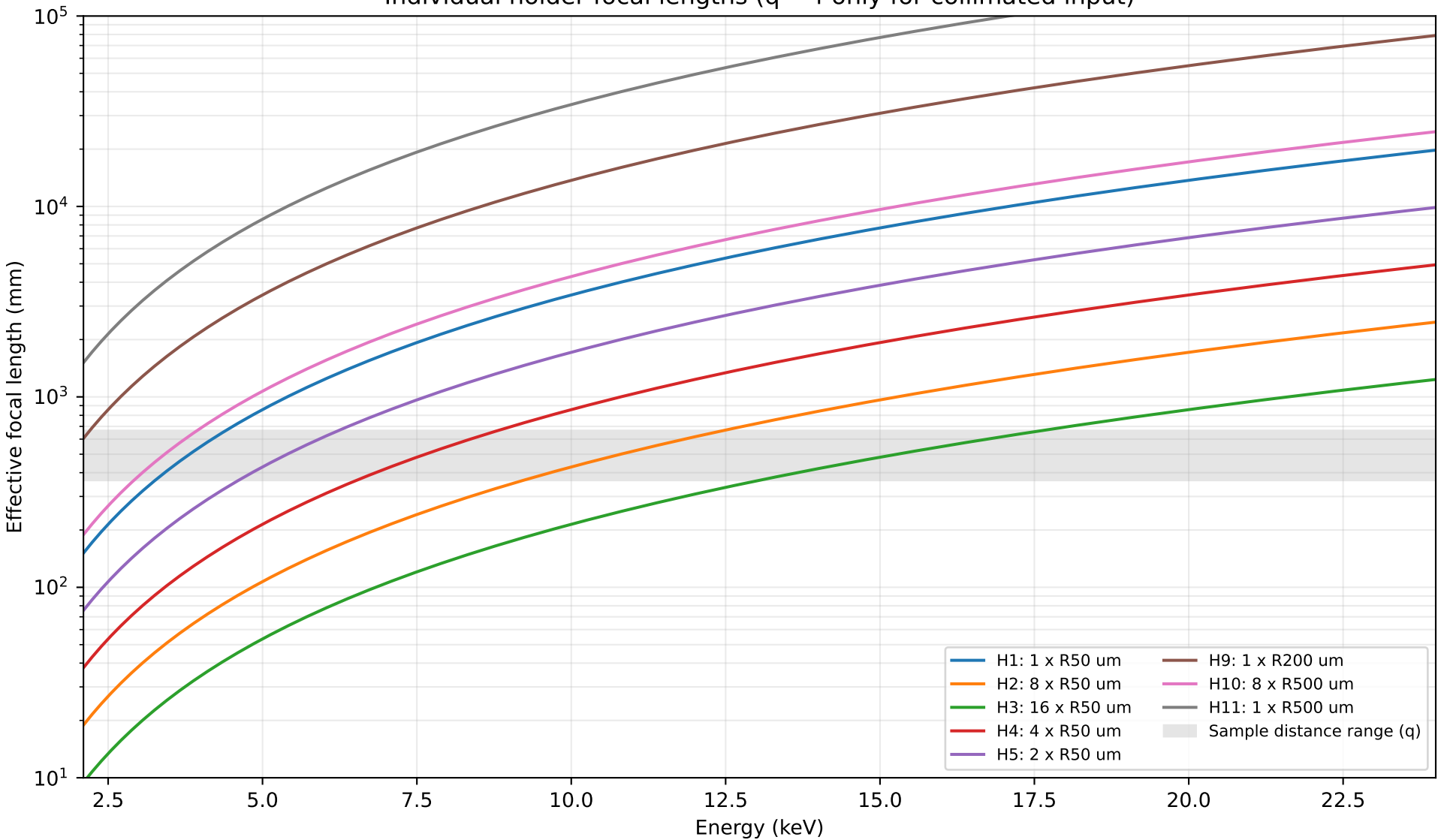
NEXT CALIBRATIONS / DECISIONS

1. Confirm material, per-surface radii, and holder axial spacing.
2. Measure at least one energy + holders + best-focus Z point.
3. Confirm actual motor travel and choose working end margins.
4. Choose alignment signal: transmission or measured spot size.
5. Decide aperture-12 behavior and persistence of IN positions.
6. Check horizontal/vertical curvature separately if astigmatic.

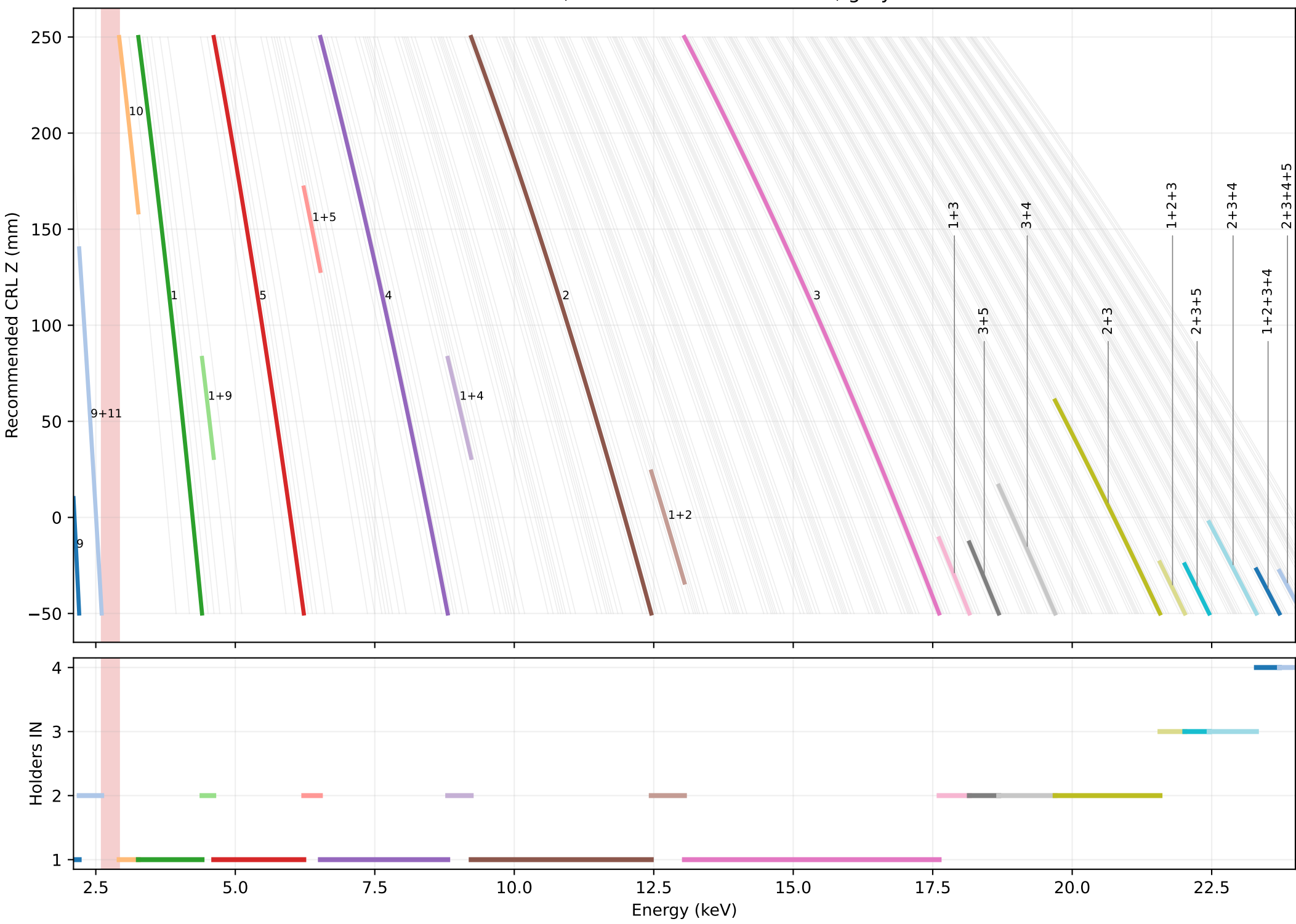
The existing lens1...lens12, x/y/z/ph/th remain motor components.

Initial CRL helpers expose inventory and offline calculations.

Individual holder focal lengths ($q = f$ only for collimated input)



Fewest-holder selection; labels are holder numbers; gray = alternatives



Recommended energy bands

Energy (keV)	Holders IN	N	Z start -> Z end (mm)
2.100- 2.202	9	1	10.1 -> -50.0
2.202- 2.605	9+11	2	140.0 -> -50.0
2.605- 2.918	UNREACHABLE		
2.918- 3.262	10	8	250.0 -> 158.8
3.262- 4.404	1	1	250.0 -> -50.0
4.404- 4.614	1+9	2	83.0 -> 31.0
4.614- 6.228	5	2	250.0 -> -50.0
6.228- 6.525	1+5	3	171.7 -> 128.3
6.525- 8.807	4	4	250.0 -> -50.0
8.807- 9.228	1+4	5	83.0 -> 31.0
9.228- 12.455	2	8	250.0 -> -50.0
12.455- 13.050	1+2	9	23.9 -> -33.9
13.050- 17.614	3	16	250.0 -> -50.0
17.614- 18.156	1+3	17	-10.9 -> -50.0
18.156- 18.683	3+5	18	-13.1 -> -50.0
18.683- 19.693	3+4	20	16.5 -> -50.0
19.693- 21.573	2+3	24	60.8 -> -50.0
21.573- 22.018	1+2+3	25	-23.4 -> -50.0
22.018- 22.454	2+3+5	26	-24.4 -> -50.0
22.454- 23.302	2+3+4	28	-2.5 -> -50.0
23.302- 23.714	1+2+3+4	29	-27.1 -> -50.0
23.714- 24.000	2+3+4+5	30	-27.8 -> -43.4

Endpoints rounded; use selection_bands.csv for precision.

At shared endpoints both adjacent choices can be reachable.

Use recommend() at the exact energy, not interpolation across a holder change. N is the total number of individual lenses.

Sensitivity to incident curvature, holding baseline lens choice fixed

